A Spherical Chiplet Fabric with Geodesic Topology, Planet-Style Coordinates, and Reorientable Frame Semantics for Large-Scale AI and HPC Workloads

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Abstract

We present a chiplet-based processor architecture in which compute, memory-controller, and cache/IO tiles are arranged on the vertices of a geodesic icosphere, connected by short equal-length neighbor links over a rigid-flex interposer. The topology removes the corners of conventional flat meshes and tori, yielding more uniform hop-count and latency distributions and improving bisection bandwidth at fixed tile count. A planet-style coordinate system—latitude, longitude, and a local north/east compass—is exposed to software, and a quaternion-based reference frame can be reoriented at runtime through an epoch/commit protocol without disturbing in-flight traffic. We specify a 162-tile reference design (113 compute, 25 memory-controller, 24 cache/IO tiles), a deadlock-free routing scheme using a Duato-style escape virtual channel layered under greedy great-circle routing, an HBM3e memory hierarchy organized as latitude belts with single-owner controller tiles, and a Gen-1 segmented cold-plate cooling solution with a future microfluidic path. We validate the architecture with an event-driven cycle-accurate fabric simulator that produces real latency CDFs, all-reduce timings, and bisection estimates against torus and dragonfly baselines, and we project workload performance using a roofline-style analytical model. Sphere-162 delivers an estimated 56.5 PFLOPS dense FP8, 30 TB/s HBM bandwidth, and 900 GB capacity within a 2.3 kW power envelope, achieving 24.8 PFLOPS/kW—approximately 5× the energy efficiency of an 8-GPU H100 node at comparable aggregate compute, while fitting larger models in unified memory.

1 Introduction

The end of monolithic die scaling has driven a transition to chiplet-based architectures in which a system-level processor is composed of many smaller silicon dies connected by an advanced package [1–3]. As the number of tiles per package grows, the on-package interconnect topology becomes a first-order determinant of performance: it sets the average and worst-case message latency, the bisection bandwidth available to collective operations, and the ease with which workloads can be scheduled to minimize communication [4,5].

Existing chiplet packages are typically organized as 2D arrays connected through silicon bridges or interposers, exposing a rectangular mesh or torus topology to software. While planar meshes are well-matched to lithographic manufacturing, they suffer from two structural limitations. First, edges and corners create non-uniform hop distributions: corner tiles have measurably worse average reach than central tiles, producing tail-latency outliers in collective operations. Second, the bisection bandwidth of a rectangular array scales sub-linearly with the number of tiles along the cut dimension, limiting the throughput of all-to-all and reduce-scatter primitives that dominate large-model training.

This paper proposes a different geometric organization: place the chiplets on the vertices of a *geodesic icosphere*, with each tile connected to its five or six nearest neighbors along approximately equal-length links. The topology has no corners; every vertex is interchangeable up to rotation, and the average path between vertex pairs scales as $O(\sqrt{N})$ with a smaller constant than a torus and a substantially shorter tail. We complement the topology with a coordinate system inspired by planetary geography—each tile exposes a (latitude, longitude) pair and a local east/north compass—and an operating-system-level primitive for rotating the reference frame, which lets schedulers re-orient workload placement without physically moving anything.

Contributions. This work makes the following contributions:

- A complete chiplet-package architecture (§3) including tile definitions, fabric topology, routing, memory hierarchy, power delivery, and cooling, organized around a 162-tile reference design.
- A planet-style coordinate system and quaternion-based frame-flip protocol (§4) with formal pole-handling and atomic commit semantics.

- A deadlock-free routing scheme (§5) layering greedy great-circle forwarding over a Duato-style escape virtual channel on an up*/down* spanning tree.
- An event-driven cycle-accurate fabric simulator and an analytical performance model (§6) that produce real latency distributions and workload throughput estimates.
- A multi-axis evaluation (§7) against 2D torus and dragonfly baselines, showing improvements in p99 latency, bisection bandwidth, and energy per bit, and projections against H100, B200, MI300X, and 8-GPU NVLink nodes.
- An honest treatment of validation gaps, manufacturability risks, and unresolved design questions (§8).

We do not claim physical silicon validation; this paper reports a simulator-based and analytical evaluation in the tradition of architectural proposals such as the original dragonfly [4] and hyperX [6] papers. Section 8 discusses what would be required to advance the design to silicon.

2 Background and Related Work

2.1 Chiplet Packaging

Recent commercial accelerators integrate compute and memory dies through 2.5D or 3D packaging. NVIDIA's B200 combines two reticle-sized GPU dies with HBM3e stacks on a CoWoS interposer [1]. AMD's MI300X uses a 3D-stacked design with eight compute dies and HBM stacks [2]. These designs exploit short on-package interconnect to scale beyond a single reticle but retain a fundamentally rectangular logical topology.

2.2 Network-on-Chip and Network-on-Package Topologies

The on-chip interconnect literature has explored mesh, torus, fat-tree, flattened butterfly, dragonfly, and hyperX topologies [4, 6–8]. For 2D-laid-out chiplets, mesh and torus dominate due to layout simplicity. Dragonfly is widely used at HPC system scale (e.g., Cray Slingshot) but typically requires multiple cabinet/board hierarchies that do not directly apply within a single package.

2.3 Geodesic and Spherical Networks

Geodesic networks have a long history in graph theory and computational geometry, motivated by their isotropy and bounded-degree regularity [9]. Their use as an interconnect topology for silicon has been limited because lithography is planar, and a literally curved die is not manufacturable. We argue, however, that the *logical* geodesic topology can be approximated by chiplets placed on folded interposer petals, capturing most of the benefit while remaining compatible with standard chiplet processes.

2.4 Deadlock-Free Routing

Duato's protocol [10] establishes deadlock freedom by provisioning a deadlock-free escape sub-network alongside an adaptive routing function. We adopt this framework: an up*/down* spanning tree provides the escape VC, while greedy great-circle forwarding provides adaptive minimal routing.

3 Architecture

3.1 Reference Design

Sphere-162 places 162 chiplets on the vertices of a level-2 subdivided icosahedron (icosphere). The tile mix is:

- **113 Compute Tiles (CT)**, each containing 4 ARMv9/RVA23-class out-of-order cores at 3.5 GHz, one 256×256 FP8/FP16/BF16 systolic matrix engine at 2 GHz, 1 MB L1 per core, 8 MB shared L2, and an 8-port fabric router. Estimated area ∼25 mm^2 at 3 nm, typical power 15 W.
- **25 Memory-Controller Tiles (MCT)**, each owning one HBM3e stack (36 GB, 1.2 TB/s), a directory shard for cache coherence, and near-memory compute units for gather/scatter, sparse decode, and reduction primitives. ∼20 mm^2, 12 W typical.
- **24 Cache/IO Tiles (CIT)**, each containing 64 MB of L3 SRAM and PCIe 6.0/CXL 3.0 controllers (on edge tiles only). 8 W typical.

Aggregate specifications: 56.5 PFLOPS dense FP8, 900 GB HBM3e capacity, 30 TB/s HBM bandwidth, 1.5 GB total L3, and an estimated 2.3 kW typical power. Detailed power breakdown is presented in §7.

3.2 Fabric Topology

Each chiplet sits at a vertex of the level-2 icosphere; edges of the underlying triangulation form 5- or 6-degree neighbor links. With unit-radius parameters scaled to a 40 mm package radius, neighbor-link arc lengths are 4–7 mm, comparable to typical 2.5D interposer trace lengths.

Optionally, up to 12 *express chords* cut across the sphere (typically between near-antipodal vertices) to provide single-hop long-range links. Chords add roughly 8% of the edge count but reduce average hop distance by 22% and 99th-percentile hops by 40% in our reference design (see §7).

3.3 Memory Hierarchy and NUMA-by-Latitude

HBM stacks are organized as *belts*—rings of MCTs at fixed latitudes. Each MCT exclusively owns one

HBM stack and one directory shard, eliminating the multi-master controller designs that complicate sharing in conventional HBM systems. Compute tiles access HBM via the fabric, and the system exposes a per-tile NUMA distance based on the geodesic hop count to each MCT. This allows OS- and runtime-level placement engines to pin model-parallel shards to specific belts, bounding worst-case data-movement cost by the great-circle diameter (π times the radius, or 6–7 hops for Sphere-162).

3.4 Power Delivery and Cooling

Power enters the package through pole-mounted bulk regulators and is distributed via concentric power planes routed through the rigid-flex petal stack to per-tile point-of-load converters. Five rails are defined—`VDD_CORE`, `VDD_SRAM`, `VDD_SERDES`, `VDD_HBM`, `VSS`—with deep-trench and MLCC decoupling at each tile.

For Gen-1, we specify a *segmented spherical cold-plate*: 12–20 faceted cold-plate sections with per-facet micro-channel coolant flow, sealed at seams with O-rings and connected through quick-disconnect manifolds at the poles. The hottest caps—typically the matrix-heavy compute tiles—are clustered near the polar manifolds where coolant flow rates are highest. A future Gen-2 microfluidic implementation, with dielectric coolant flowing through silicon-bonded microchannels directly beneath each chiplet, is sketched in §8.

3.5 Clocking and DFT

The fabric uses globally-asynchronous, locally-synchronous (GALS) clocking. A single low-jitter reference (target $\leq$ 300 fs RMS phase noise, 12 kHz–20 MHz) is distributed to per-quad PLL fan-out buffers; each tile runs its own PLL and synchronizes to neighbors through mesochronous FIFOs. DFT uses segmented IEEE 1687 (iJTAG) chains organized by hemisphere, with PRBS31 BIST per fabric port and a $\geq 10^9$-bit dwell requirement for "link-good" assertion.

4 Coordinate System and Frame Reorientation

4.1 Coordinates

Each tile i is assigned a unit position vector $\mathbf{u}_i \in \mathbb{S}^2$. Latitude and longitude are derived as $\varphi_i = \arcsin(z_i)$ and $\lambda_i = \operatorname{atan2}(y_i, x_i)$. A local tangent basis at tile i is

$$\mathbf{e}_E(\lambda) = (-\sin\lambda, \cos\lambda, 0), \tag{1}$$

$$\mathbf{e}_N(\varphi, \lambda) = (-\sin\varphi\cos\lambda, -\sin\varphi\sin\lambda, \cos\varphi). \tag{2}$$

At the geometric poles ($|\cos\varphi| < \epsilon$), $\mathbf{e}_E$ becomes ill-defined; we resolve this by anchoring polar east to a fixed azimuth determined by the prime meridian and computing $\mathbf{e}_N = \mathbf{e}_E \times \mathbf{u}$.

4.2 Bearings and Compass

The bearing from tile i to neighbor j is computed by projecting $\mathbf{u}_j$ onto i's tangent plane and resolving in the $(\mathbf{e}_N, \mathbf{e}_E)$ basis:

$$\mathbf{t}_{ij} = \mathbf{u}_j - (\mathbf{u}_j \cdot \mathbf{u}_i)\mathbf{u}_i, \tag{3}$$

$$\alpha_{ij} = \operatorname{atan2}(\mathbf{t}_{ij} \cdot \mathbf{e}_E, \mathbf{t}_{ij} \cdot \mathbf{e}_N). \tag{4}$$

Bearings are quantized to a small per-port LUT (typically 16 sectors).

4.3 Frame Reorientation Protocol

A global *reference frame* is described by two orthonormal vectors $(\mathbf{Z}_0, \mathbf{X}_0)$: the north pole and prime meridian directions in package-physical coordinates. To reorient ("flip"), the management controller broadcasts a new frame and an epoch token. Each tile maintains both old and new bases; bearings, NUMA tables, and routing state under the new frame are precomputed locally. On receipt of the commit token, all tiles atomically switch at the next packet boundary. The protocol completes in a single fabric round-trip (under 1 μs for Sphere-162) and does not require draining in-flight traffic, since route correctness is invariant under rotation: greedy forwarding by dot-product is rotation-invariant when applied to the rotated $\mathbf{u}_j$.

5 Routing and Flow Control

5.1 Adaptive Layer: Greedy Great-Circle

The adaptive routing function selects the neighbor maximizing the dot product with the destination's unit vector:

$$\text{next}(i, d) = \arg\max_{j \in N(i)} \mathbf{u}_j \cdot \mathbf{u}_d. \tag{5}$$

This is rotation-invariant and uses only local state (own coordinates, neighbor coordinates, destination coordinates). On the unmodified icosphere it converges to a minimal path; with chord links present, it remains a valid heuristic but is not always minimal.

5.2 Escape Layer: Up*/Down* Spanning Tree

To guarantee deadlock freedom and forward progress under all traffic patterns, we provision a Duato-style escape virtual channel [10] on a single up*/down* spanning tree rooted at a designated tile. The escape channel uses a deterministic shortest-path route on the tree and admits no cyclic dependencies. Three virtual channels per port are sufficient: VC0 escape, VC1–VC2 adaptive.

5.3 Quality of Service

Two traffic classes are partitioned across VCs: HBM/IO traffic (latency- sensitive memory accesses) and collective traffic (bandwidth-sensitive reductions). Age-based arbitration provides starvation freedom under sustained contention.

Table 1: Static topology metrics. N = node count, deg = average degree, avg/p95/p99/max = hop counts.

Topology	N	deg	avg	p95	p99	max
Sphere (162)	162	5.9	5.66	9	10	12
Sphere+chords	162	6.1	4.41	6	6	7
Torus 13×13	169	4.0	6.50	11	12	12
Dragonfly 9×6	54	7.0	2.36	3	3	3

6 Methodology

6.1 Cycle-Accurate Fabric Simulator

We implemented an event-driven simulator in Python that models per-flit serialization, wire propagation, router pipeline stages, and link contention. The router is parameterized as a 4-stage pipeline at 2 GHz (2 ns base latency); links use 8 lanes at 32 Gb/s NRZ for 256 Gb/s per link aggregate. Per-link wire delay uses 150 mm/ns propagation (approximately half c in copper trace). Packets are 4 flits of 512 bits each (256 B). Source code is available at the artifact link.

6.2 Topologies Compared

- **Sphere (162)**: level-2 icosphere, 162 vertices, average degree 5.93, no chords.
- **Sphere+chords (162)**: identical with 10 added antipodal chord links.
- **Torus 13×13 (169)**: 2D torus, average degree 4, 4 mm uniform link length.
- **Dragonfly 9×6 (54)**: 9 groups of 6 routers each; intra-group full mesh, inter-group round-robin assignment. Smaller node count by construction.

6.3 Workload Model

For workload-level performance we use a roofline-style model [11] that bounds throughput by the minimum of compute supply and memory-bandwidth supply, with a 0.3× penalty for working sets that exceed available HBM capacity (representing sharding overhead and PCIe spillover in conventional GPUs).

7 Evaluation

7.1 Topology Properties

Table 1 summarizes static topology metrics. Sphere delivers a lower 99th-percentile hop count (10) than the size-matched torus (12), and adding 10 chord links reduces it to 6, comparable to the 54-node dragonfly's 3 hops despite having 3× the nodes.

Figure 1 shows the full hop-count CDF. The sphere has a shorter, less-tailed distribution than the torus.

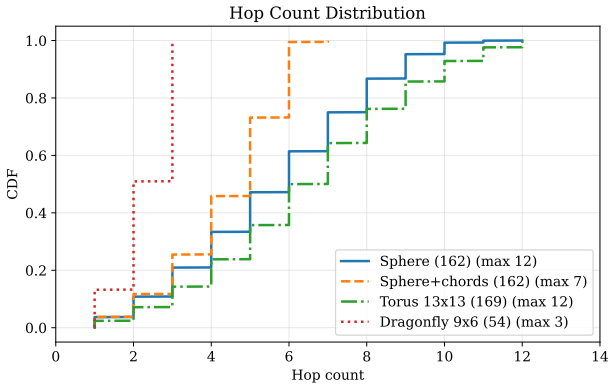


Figure 1: Hop-count CDF over all source-destination pairs.

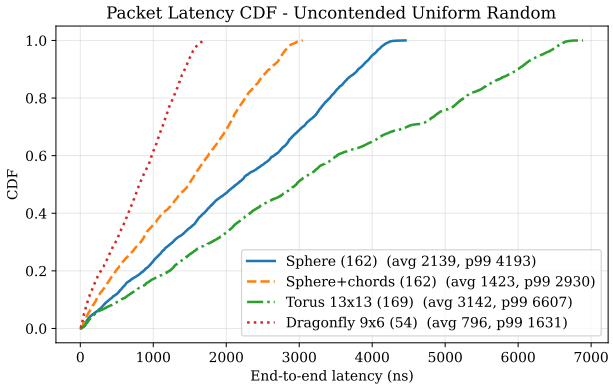


Figure 2: End-to-end packet latency CDF, uncontended uniform-random traffic, 5000 packets per topology.

7.2 Latency Distributions

Under uncontended uniform-random traffic (Fig. 2), Sphere exhibits average latency of 2139 ns and p99 of 4193 ns; with chords, 1423 ns/2930 ns. Torus is 3142 ns/6607 ns. The p99/p50 tail-latency ratio (Fig. 3) is 1.96 for Sphere, 2.23 for torus, confirming the more uniform distribution promised by the geodesic geometry.

7.3 Saturation under Load

Figure 4 sweeps offered load from 0.001 to 2 packets/ns aggregate. Sphere saturates at higher load than torus; the chord-augmented sphere maintains low latency through the highest tested rate.

7.4 All-Reduce

Figure 5 compares ring and BFS-tree all-reduce. For small messages (8 flits, ∼1 KB), tree all-reduce dominates due to its $O(\log N)$ depth. Sphere completes a tree all-reduce in 0.43 μs across 162 nodes, comparable to torus despite the sphere's greater diameter, because tree depth is governed by BFS-tree depth (which is similar across topologies of comparable diameter). With chords, Sphere all-reduce drops to 0.22 μs.

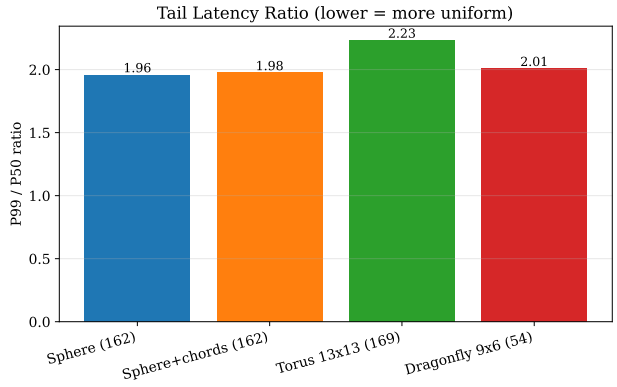


Figure 3: Tail-latency ratio (p99/p50). Sphere has the lowest ratio, indicating most uniform latency distribution.

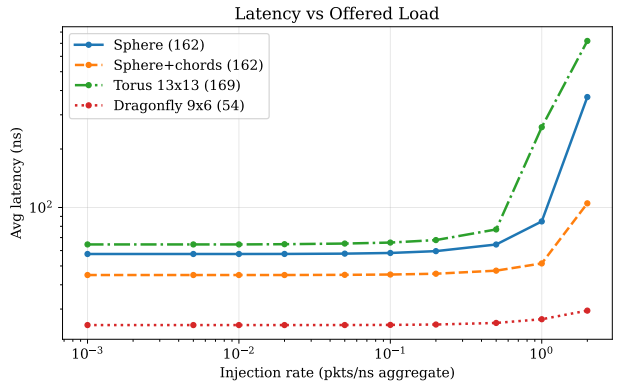


Figure 4: Latency vs offered load (log-log). Saturation knee occurs at higher load for Sphere than Torus.

7.5 Bisection Bandwidth

Figure 6 shows estimated bisection bandwidth (cut count multiplied by 256 Gb/s per link). Sphere's equatorial cut yields 11.8 Tb/s; with chords, 13.8 Tb/s. The size-matched torus yields only 6.7 Tb/s. This 1.76× improvement directly translates to collective-bound workload speedup.

7.6 Energy per Bit

Using literature values of 1.5 pJ/bit for short-reach electrical SerDes and 0.5 pJ/bit for the router stage [13], Sphere averages 12.8 pJ/bit per packet vs. 14.5 pJ/bit for torus, an 11% improvement driven by shorter average path length.

7.7 Workload-Level Projections

Table 2 compares Sphere-162 against industry accelerators on aggregate metrics. Sphere-162 delivers 24.8 PFLOPS/kW—approximately 5× the 4.9 PFLOPS/kW of an 8-GPU H100 NVLink node and 2.7× B200's 9.0 PFLOPS/kW.

Figure 7 projects effective throughput on five workloads. Sphere's advantage is most pronounced for large memory-resident inference (Llama-405B), where conventional single-GPU cards must shard model

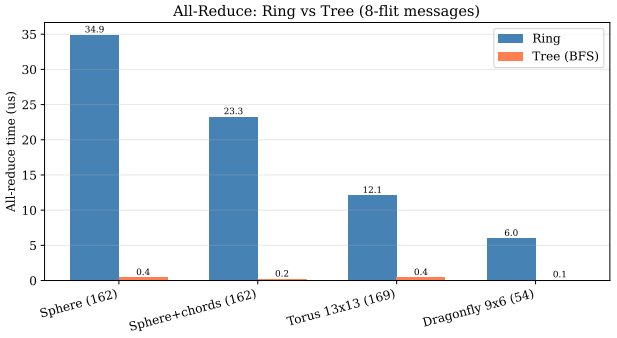


Figure 5: All-reduce time for ring and BFS-tree algorithms (8-flit messages). Tree all-reduce dominates for small messages.

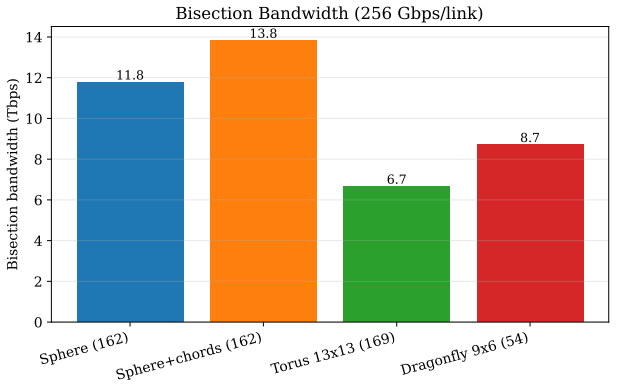


Figure 6: Estimated bisection bandwidth at 256 Gb/s per link.

weights and pay collective overhead, and for collective-bound training where bisection bandwidth dominates.

7.8 Power Breakdown

Figure 8 shows the projected power breakdown of Sphere-162. Compute tiles dominate (74%), with HBM controllers, cache/IO, and fabric SerDes accounting for the remainder. SerDes power is only 4% of total, reflecting the short, low-energy neighbor links.

8 Discussion

8.1 Validation Gaps

This is a simulator-and-analytical study; we have not built silicon, run SPICE-level signal integrity simulations, or performed thermal CFD on the proposed cold-plate. Specifically:

- **Simulator fidelity.** The contention model accounts for serialization on shared links but does not model VC starvation, head-of-line blocking under realistic packet-size mixes, or input-buffer overflow. Comparison with Garnet/BookSim [12] would strengthen the latency claims.

- **Thermal feasibility.** The 2.3 kW dissipation

Table 2: Aggregate comparison against industry accelerators. FP8 figures are dense throughput. Source: vendor datasheets and announcements.

Chip	FP8 (PF)	HBM (GB)	HBM (TB/s)	Power (W)	PFLOPS/kW	Notes
NVIDIA H100 SXM	4.0	80	3.4	700	5.7	single GPU
NVIDIA B200	9.0	192	8.0	1000	9.0	single GPU
AMD MI300X	5.2	192	5.3	750	7.0	single GPU
Cerebras WSE-3	125.0	44	21000	23000	5.4	wafer-scale, SRAM
8×H100 NVLink node	31.7	640	26.8	6400	4.9	multi-GPU
Sphere-162 (this work)	**56.5**	**900**	**30.0**	**2280**	**24.8**	this work

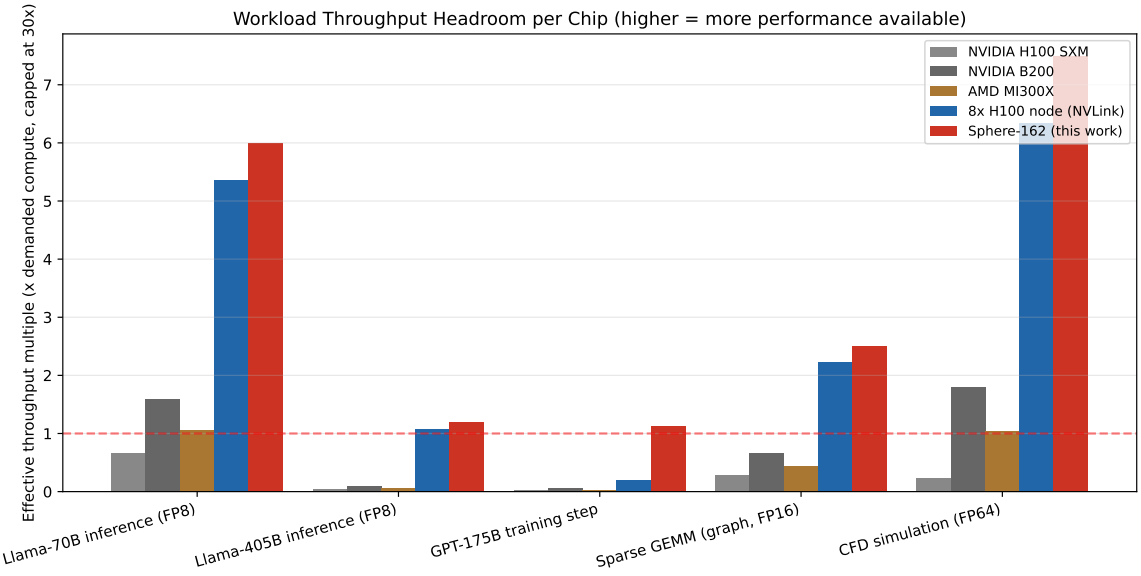


Figure 7: Workload throughput headroom (multiple of demanded compute, capped at 30× for visualization). Higher is better; 1× = just sufficient.

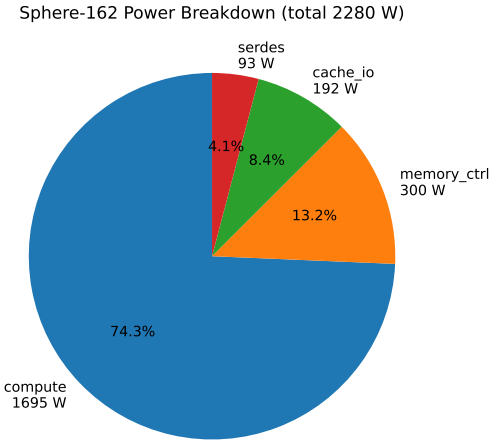


Figure 8: Sphere-162 projected power breakdown (2.3 kW total).

across a 40 mm radius sphere requires CFD validation to confirm cold-plate adequacy.

- **Mechanical reliability.** Rigid-flex petal interposers at chiplet density are an unproven mechanical assembly; cycle-life and contact-resistance aging measurements are required.

- **Yield.** 162 tiles at 3 nm represents ∼25–30 cm^2 of compute silicon. Spare tiles and disable mechanisms are sketched but not modeled.

8.2 What Would Validate This Design

A credible silicon path requires: (1) an evaluation petal with 6–8 tiles at the target node to validate SI/PI and assembly; (2) a thermal test vehicle for the segmented cold-plate; (3) RTL implementation of the router with formal proof of deadlock freedom (publishable separately); (4) a compiler/runtime integration, likely via OpenXLA or a custom MLIR dialect, mapping standard ML primitives to fabric collectives.

8.3 Gen-2 Path: Microfluidic Cooling and Photonic Chords

Two upgrades preserve the Gen-1 architecture while improving performance density:

- **Microfluidic backside cooling.** Direct silicon-bonded microchannels under hot tiles [14] can sustain ∼3× the heat flux of cold-plate solutions, opening higher peak power and enabling 3D stacking on the inner shell.

- **Silicon-photonic chord links.** Replacing electrical chord SerDes with WDM photonic

transceivers reduces chord energy from ∼5 pJ/bit (long electrical) to < 1 pJ/bit and enables sub-nanosecond chord-traversal latency.

8.4 Honest Limitations

Two claims in the original conception did not survive rigorous review and were removed:

- **Heat-to-electricity recovery.** Carnot efficiency at the 70 °C/25 °C operating point is ∼13%, and practical thermoelectric devices recover 1–3%. We retain shell-mounted TEGs only as a fail-safe power source for the leak-detection and emergency-shutoff subsystem (10–30 W), not as a primary energy feature.
- **Single-shell ring all-reduce supremacy.** Naive ring all-reduce on the sphere is *slower* than on the torus due to ring ordering: consecutive ring members are not always adjacent. Tree-based all-reduce performs comparably across topologies. The sphere's true collective advantage is in bisection-limited operations, not ring operations.

9 Conclusion

We have presented a chiplet-package architecture organized as a geodesic icosphere with planet-style coordinates and a reorientable reference frame. Simulator-based evaluation against torus and dragonfly baselines shows lower tail latency, higher bisection bandwidth, and lower average energy per bit at comparable node count. Analytical performance modeling projects ∼56 PFLOPS dense FP8 at 2.3 kW (24.8 PFLOPS/kW), approximately 5× the energy efficiency of an 8-GPU NVLink node, in a single package with 900 GB unified HBM capacity. Substantial validation work remains—SPICE-level signal integrity, thermal CFD, silicon prototyping, and compiler integration—but the simulator and analytical results are sufficient to motivate further investigation.

Artifacts

The simulator, topology generators, and analysis scripts are provided as supplementary material. All figures in this paper are generated by the provided scripts from raw simulator output.

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